Unit 6: Ecology

Complete Study Notes – Grade 9 Biology

6.1 Ecology

Every day you share your environment with many organisms. These can be as small as microbes or as large as elephants. You need to know about your environment because it affects you and every other organism in it.

6.1.1 Definitions of Common Ecological Terms

Ecology is the branch of biology that deals with the study of the interactions among organisms and with their environment. Scientists who study ecology are called **ecologists**.

An **ecosystem** consists of the physical environment (abiotic factors) and all the living things (biotic factors) within it.

Key terms:

- **Habitat –** the place where an organism lives (e.g., a pond, a forest).
- **Niche –** the role an organism plays in its ecosystem, including its habitat, resources, and interactions.
- **Population –** a group of individuals of the same species living in a specific area.
- **Community** – all the different populations of organisms living together in a specific area.
- **Ecosystem –** the community plus the non-living (abiotic) environment.
- **Biosphere –** the part of the Earth that supports life (sum of all ecosystems).

Ecologists use both qualitative and quantitative research. They observe organisms and their environment, and they make measurements and do experiments.

6.1.2 Biotic and Abiotic Components

Abiotic factors are non-living components of the ecosystem that influence the distributions of organisms. Examples include:

- **Solar energy** – powers most ecosystems through photosynthesis.
- **Temperature –** affects metabolism; most organisms function best within a specific range.
- **Water –** essential for all life; terrestrial organisms face drying out.
- **Nutrients –** inorganic nutrients (nitrogen, phosphorus) limit plant growth.
- **Salinity** – important in aquatic ecosystems.
- **Oxygen** – dissolved oxygen critical for aquatic organisms.
- **Wind –** affects evaporation, water loss, and can cause wind chill.
- **Natural disturbances –** storms, fires, volcanic eruptions shape ecosystems.

Biotic factors are living components that influence the distributions of organisms. Examples include:

- **Predators –** organisms that kill and eat other organisms.
- **Herbivores** – organisms that eat plants or algae.
- **Parasites** – organisms that live on or in a host and harm it.
- **Pathogens** – disease-causing organisms.
- **Competitors –** organisms that compete for the same resources.
- **Pollinators** – organisms that help plants reproduce.

Example: In a grassland, temperature and rainfall (abiotic) determine the types of grasses that grow. The presence of grazing animals (biotic) affects the distribution of plant species.

6.1.3 Ecological Levels

Ecologists study living world at different levels of organization, from individual organisms to the biosphere.

Organism – a single individual.

Population – a group of the same species living in the same area.

Community – all the different populations living together.

Ecosystem – the community plus the abiotic environment.

Biosphere – the global sum of all ecosystems.

A **population** consists of interbreeding organisms of the same species. Populations may compete for resources. A **biological community** consists of different species interacting (predation, parasitism, competition, pollination).

Methods for studying populations:

- **Quadrats –** square frames used to sample immobile organisms (plants, slow-moving animals). Count individuals within the quadrat.
- **Mark-recapture –** for mobile animals; capture, mark, release, then recapture later to estimate population size.

6.1.4 Ecosystems

An **ecosystem** is a community of organisms in a habitat, plus the non-living part of the environment. Examples: a lake, a woodland, a desert.

Ecosystems are **self-supporting**: plants absorb light and rainwater for photosynthesis; animals feed on plants and each other; dead remains are decomposed by fungi and bacteria, returning nutrients to the soil.

Types of ecosystems:

- **Terrestrial ecosystems –** on land (forests, grasslands, deserts).
- **Aquatic ecosystems –** in water (freshwater: ponds, lakes, rivers; marine: oceans, estuaries).

Example – Lake ecosystem: The plant and animal communities, plus water, minerals, dissolved oxygen, soil, and sunlight, make up the lake ecosystem. Materials are recycled within the ecosystem, and energy flows from the Sun through producers to consumers.

6.1.5 Biomes

A **biome** is a major terrestrial or aquatic life zone characterized by vegetation type (terrestrial) or physical environment (aquatic).

Terrestrial Biomes:

Tropical Rainforest

- Location: near equator (10° N and S), South America, Africa, Southeast Asia.
- Climate: year-round rain (130–200 cm), average 25°C.
- Vegetation: broadleaf evergreen trees, vertically layered, intense competition for light.
- Animals: high biodiversity (amphibians, birds, mammals, insects).
- Threats: deforestation for lumber, agriculture; reduces CO_2 absorption.

Savanna

- Location: equatorial and sub-equatorial regions (Africa, India, Australia).
- Climate: 30–50 cm rain/year; dry season 8-9 months; warm year-round (24–29°C).
- Vegetation: scattered trees, fire-adapted grasses; tolerant of grazing.
- Animals: giraffes, zebras, elephants, antelopes, lions, hyenas.
- Ethiopia: has extensive savanna areas (e.g., in the lowlands).

Desert

- Location: 30° N and S latitude (Sahara, Gobi, Atacama).
- Climate: < 10 cm rain/year; hot deserts > 50°C, cold deserts < -30°C.
- Vegetation: succulents (cacti), deeply rooted shrubs, herbs that grow after rain.
- Animals: reptiles, scorpions, ants, nocturnal rodents; adaptations for water conservation.

Temperate Grassland

- Location: North American prairies, Eurasian steppes.
- Climate: warm summers, cold winters; 25–100 cm rain/year.
- Vegetation: grasses with deep roots; rich topsoil.
- Animals: herds of bison, pronghorn antelope, wolves; now mostly converted to farmland.

Boreal Forest (Taiga)

- Location: south of Arctic Circle (Canada, Alaska, Russia, northern Europe).
- Climate: cold, dry winters; short, cool, wet summers; 40–100 cm precipitation (mostly snow).
- Vegetation: cone-bearing conifers (pines, spruces) – needle-like leaves, evergreen.
- Animals: mammals, birds; many hibernate or migrate.

Temperate Broadleaf Forest

- Location: mid-latitudes Northern Hemisphere, also Chile, South Africa, Australia.
- Climate: 70–200+ cm rain/year; hot humid summers, winter average 0°C.
- Vegetation: deciduous trees (drop leaves in winter); many mammals hibernate, birds migrate.
- Human impact: heavily settled, but forests are recovering in some areas.

Tundra

- Location: Arctic regions (20% of Earth's land), also alpine tundra on mountains.
- Climate: 20–60 cm precipitation; very cold winters (< -30°C), summer avg < 10°C.
- Vegetation: herbaceous (mosses, grasses, dwarf shrubs, lichens); permafrost layer.
- Animals: caribou, musk oxen, arctic foxes, migratory birds.

Global biomes map: Temperature and precipitation are key determinants of biome distribution.

Aquatic Biomes

Aquatic biomes occupy roughly 75% of Earth's surface. They are classified by salinity:

- **Freshwater biomes –** salt concentration < 1%: lakes, streams/rivers, wetlands.
- **Marine biomes –** salt concentration ~3%: oceans, intertidal zones, coral reefs, estuaries.

Lakes: Standing water bodies, stratified into zones: *photic* (light penetrates), *aphotic* (no light), *pelagic* (open water), *benthic* (bottom). Phytoplankton are primary producers. Eutrophication from nutrient runoff causes algal blooms and oxygen depletion.

Wetlands: Areas saturated with water (marshes, swamps, bogs). High productivity; filter pollutants; reduce flooding. Destroyed by draining and filling.

Streams and Rivers: Headwaters are cold, clear, swift; lower reaches are warmer, turbid. Support diverse fish and invertebrates; dams and pollution degrade water quality.

Estuaries: Transition between river and sea (where fresh and salt water mix). Very productive; breeding and feeding grounds for many species. Threatened by filling, dredging, pollution.

Oceans: Major marine biome. Zones: intertidal (between high and low tide), neritic (shallow coastal), oceanic (deep sea). Coral reefs are highly biodiverse, but threatened by warming and acidification.

6.1.6 Ecological Succession

Ecological succession is the process by which the composition of a community changes over time after a disturbance.

Primary succession: Begins in a virtually lifeless area (e.g., new volcanic island, glacier retreat). Pioneer species (lichens, mosses) colonize bare rock. They weather the rock, and organic matter accumulates, forming soil. Over hundreds to thousands of years, a climax community (stable, mature) develops.

Secondary succession: Occurs after a disturbance that removes most but not all of the community (e.g., fire, farming). Soil is still present. The area may return to its original state over time. Example: abandoned farmland – first weeds, then shrubs, then forest.

Key processes: Early species may facilitate (help) later species, inhibit them, or tolerate them.

6.2 Ecological Relationships

Species interact in communities. These interspecific interactions can be summarized as:

Competition (–/–)

Interaction where individuals of different (or same) species use a limited resource, harming both. Example: weeds competing with crops for nutrients and water.

Predation (+/–)

One organism (predator) kills and eats another (prey). Example: lion eating zebra. Adaptations in both predators and prey evolve.

Herbivory (+/–)

An organism (herbivore) eats parts of a plant or alga, harming it. Example: cattle grazing grass; insects eating leaves.

Parasitism (+/–)

One organism (parasite) lives on or in a host, deriving nutrients while harming the host. Example: tapeworms in animals, ticks on dogs.

Mutualism (+/+)

Interaction that benefits both species. Example: flowers and their pollinators (bees get nectar, plants get pollinated); nitrogen-fixing bacteria in legume roots; mycorrhizae (fungi with plant roots).

Commensalism (+/0)

One species benefits, the other is neither helped nor harmed. Example: cattle egrets feeding on insects flushed by grazing herbivores.

Unit 6 Summary

- **Ecology** is the study of interactions between organisms and their environment.
- **Abiotic factors** (non-living) and **biotic factors** (living) determine species distributions.
- Ecological levels: organism → population → community → ecosystem → biosphere.
- An **ecosystem** consists of the community and its abiotic environment; it is self-supporting.
- **Biomes** are major life zones: tropical rainforest, savanna, desert, temperate grassland, boreal forest, temperate forest, tundra, and aquatic biomes.
- **Ecological succession** is the gradual change in community composition: primary (new land) and secondary (after disturbance).
- **Ecological relationships** include competition, predation, herbivory, parasitism, mutualism, and commensalism.
- Each interaction has a characteristic effect on the populations involved (+ beneficial, – harmful, 0 neutral).

End of Unit 6 – Ecology